1 Submersions, Immersions and Embedings

1.0.1 Local Diffeomorphism

Definition 1.0.1

If M and N are smooth manifolds with or without boundary, a map $F : M \to N$ is called a ***local diffeomorphism*** if every point $p \in M$ has a neighborhood U such that $F(U)$ is open in N and $F|_U : U \to F(U)$ is a diffeomorphism.

Theorem 1.0.2 ▸ Inverse Function Theorem for Manifolds

Suppose M and N are smooth manifolds, and $F : M \to N$ is a smooth map. If $p \in M$ is a point such that dF_p is invertible, then there are connected neighborhoods U_0 of p and V_0 of $F(p)$ such that $F|_{U_0} : U_0 \to V_0$ is a diffeomorphism.

Note.
也就是说，光滑映射在微分映射可逆的点附近是微分同胚.